Name – Kush Anil Yarwar

Reflection: Dockerizing the QR Code Generator Application

Experience:

This assignment gave me hands-on experience with Docker, from writing a Dockerfile to building an image, running a container, and pushing it to DockerHub. I also set up a GitHub Actions workflow to automate the entire build and push process whenever code is pushed to the main branch.

What I Learned:

1. How Docker containers package an application and its dependencies into a portable, isolated environment
2. How to write a Dockerfile using best practices like layered caching (copying requirements.txt before the rest of the code) and minimal base images (python:3.12-slim-bullseye)
3. How to use GitHub Actions to automate Docker image builds and pushes to DockerHub using secrets for secure credential management

Challenges Faced:

1. Docker Desktop was not running initially, which caused connection errors when trying to build the image. Starting Docker Desktop resolved this.
2. The GitHub Actions workflow failed on the first run because the DOCKERHUB_USERNAME and DOCKERHUB_TOKEN secrets had not been added to the repository yet. Once the secrets were configured under repository settings, the workflow ran successfully.

Security Considerations:

1. A non-root user (myuser) was created in the Dockerfile to run the application, reducing the risk of privilege escalation inside the container
2. The python:3.12-slim-bullseye base image minimizes the attack surface by including only essential packages
3. DockerHub credentials were stored as GitHub Secrets rather than hardcoded in the workflow file

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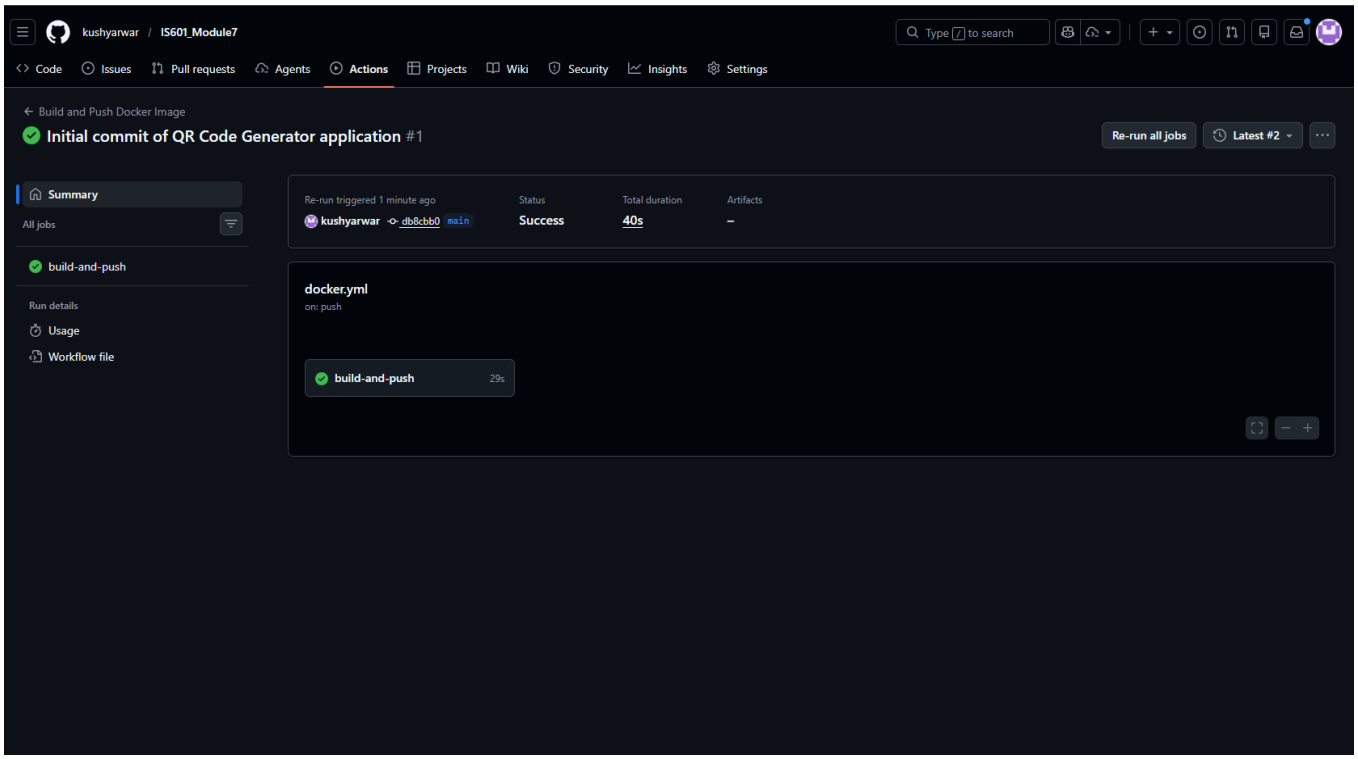
Container Logs:

```
PS C:\Users\Kush Yarwar\Desktop\Python_Web_API\IS601_Module7> docker logs qr-generator
2026-03-10 17:28:04,058 [INFO] Generating QR code for URL: http://github.com/kaw393939
2026-03-10 17:28:04,072 [INFO] QR code saved to: qr_codes/qr_20260310_172804.png
2026-03-10 17:28:04,072 [INFO] Done! QR code generated successfully.
```

GitHub Actions Workflow: